

SECURITY OPERATIONS CENTER — HOME LAB

Phase 2

Routed & Segmented Network

When a router breaks the sensor's view

The flat network is gone — a VyOS router now sits between the analyst and the attack.

What changed from Phase 1

Phase 1 — Flat

- One L2 segment (10.10.10.0/24)
- No router, no boundaries
- One tap sees ALL traffic
- Sensor placement = irrelevant
- L2 attacks (ARP) possible

Phase 2 — Routed

- Two subnets + VyOS router
- Traffic crosses only via the router
- A tap sees ONLY its segment
- Sensor placement = a decision
- L2 attacks stop at the boundary

The routed lab — real addressing

VLAN 10 (trusted / internal)

10.10.10.0/24

wazuh .20

zabbix .22

velociraptor .23

grafana .24

client .50

VyOS

.10.1 <-> .20.1

router

VLAN 20 (exposed / target)

10.10.20.0/24

attacker .40

server .60

honeypot .70

|

SENSOR tap (enp0s8)

Suricata + Zeek, promisc

Management (out-of-band): 192.168.56.0/24 — SOC admin, unchanged

Sensor placement is now a decision

A sensor sees only the segment it sits on.

Wrong: sensor on the client side

It sees the attacker doing... nothing. And nothing looks exactly like safety.

Our choice: sensor on VLAN 20 (the exposed side)

It captures every hostile action at its source — where the attacker and honeypot live.

Pure routing preserves the truth

```
attacker 10.10.20.40
  |
  VyOS router (routes, does NOT rewrite addresses)
  |
victim 10.10.10.50
```

Wazuh auth.log / Suricata alert still read:
SRC = 10.10.20.40 (the real attacker)

NO NAT

ATTRIBUTION INTACT

logs name the attacker,
not the router

What we actually built

1 VyOS router VM

one new machine — 2 NICs, one per VLAN

2 Two internal networks

vlan10 + vlan20 in VirtualBox

3 Rewire guest NICs

move each VM to its segment

4 Configure routing

eth0 .10.1 / eth1 .20.1, no NAT

5 Persist to netplan

runtime ip -> permanent config

6 Reposition the sensor

tap moved to the VLAN 20 wire

The bug that taught the most

Symptom: ping worked one way, failed the other.

A leftover 95-flat.yaml (Phase 1 config) gave the honeypot a SECOND, stale address on its VLAN 20 interface.

-> the kernel thought 10.10.10.0/24 was directly attached,
so replies to the SOC went out the WRONG wire and vanished.

Fix: remove the old phase's netplan file, drop the stale IP.

Lesson: one-way connectivity = a return-path problem. When you re-address for a new phase, REMOVE the old config — don't just add the new one.

All 5 planes — verified across the router

- | | | |
|-----------------------|----------------------------------|----------------------|
| ● Network | Suricata + Zeek tap VLAN 20 | full scan captured |
| ● Endpoint | Wazuh agents report via router | rule 5712 fired |
| ● Availability | Zabbix polls honeypot via router | ZBX green in Grafana |
| ● Forensic | Velociraptor clients connected | both segments |
| ● Deception | Cowrie honeypot on VLAN 20 | brute-force logged |

The scan, captured on VLAN 20

65,535

ports scanned

131,066

REJ (closed probed)

5

SF (open found)

```
attacker 10.10.20.40 -> honeypot 10.10.20.70 (nmap -sV -p-)
```

```
Zeek conn.log: one source, 65535 distinct ports, seconds apart
```

```
That shape IS a port scan – the sensor on VLAN 20 saw all of it.
```

Detection across the router

```
attacker (VLAN 20) brute-forces honeypot :2022
|
honeypot auth.log -> Wazuh agent
| (agent traffic ROUTES via VyOS to VLAN 10)
Wazuh manager 10.10.10.20
|
Rule 5712 (level 10) SSH brute force
SRC = 10.10.20.40 Target = honeypot
```

5712

FIRED VIA ROUTER

agent -> manager across
the segment boundary

Watching a host beyond the router

- Installed Zabbix agent on the honeypot (VLAN 20)
- Agent points at the server 10.10.10.22 (VLAN 10)
- Zabbix server polls it on :10050 — THROUGH VyOS
- Host registered, ZBX indicator green, data flowing
- Surfaced in Grafana — one pane, across the boundary

VLAN 10
watches
VLAN 20
through the router

Phase 2 — PASSED

- VyOS router deployed — two segmented VLANs
- Sensor placement made a deliberate decision (VLAN 20)
- Pure routing — attribution preserved, no NAT
- All 5 detection planes verified across the router
- Agents + Zabbix monitor the exposed segment remotely
- Config persisted to netplan, snapshot: phase2-final

What comes next?

- 1. Phase 3 — a firewall (OPNsense) with NAT
- 2. NAT rewrites the source — attribution starts to break
- 3. Recover the real attacker via the firewall's own logs
- 4. Correlation across planes becomes essential

Flat > [Routed] > Firewallled + NAT > Two firewalls + VPN

Phase 2 proved the SOC survives a router. Phase 3 attacks the thing it relies on most: attribution.